

From Brainwaves to Csound: An Exploratory Audiovisual System with EEG and LLM-Based Agent Control in Unity

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Abstract. We present an exploratory audiovisual biofeedback system that integrates real-time electroencephalogram (EEG) signals, large language model (LLM)-based assistance, and Csound within Unity, a real-time game engine, transforming a listener’s brain activity into continuously evolving sound and a synchronized visual scene. NeurOSC, a Python-based EEG acquisition and signal-processing layer, estimates power within five predefined EEG frequency bands, calculates their cross-channel mean relative composition, and sends these features to Unity via Open Sound Control (OSC). Unity converts the features into Z-scores using a calibration reference established for the current session and assigns one of five system-defined state labels: *Focused*, *Relaxed*, *Aroused*, *Drowsy*, or *Neutral*. The system uses an LLM in two roles. Before a session, a mapping assistant combines the creator’s natural-language intent with a catalog of Csound parameters parsed from the instrument’s Cabbage header and proposes EEG-to-Csound bindings that the creator can audition and revise. At run time, a scene agent responds to confirmed state transitions and other configured events. It sets larger-scale base targets for selected Csound parameters and issues coordinated visual actions, while a continuous path maps changing EEG features to smaller-scale modulation around those targets. The prototype runs in real time with a Muse S Athena headset and includes a simulator for development without hardware. Its contributions are a human-in-the-loop mapping workflow that combines parameter auto-discovery with LLM-generated suggestions, and a two-path architecture that combines continuous physiological modulation with discrete, context-informed audiovisual changes.

Keywords: Csound, CsoundUnity, EEG biofeedback, real-time audiovisual systems, human-AI collaboration

1 Introduction

Turning biofeedback into sound has a long history in computer music, from early performances to recent brain-computer interface (BCI) instruments [1]. EEG provides continuous physiological signals that have been studied in relation to affective states, including the roles of different frequency bands and channels [2]. Consumer headsets have made these signals increasingly accessible. Csound, with its mature real-time synthesis, large opcode library, and established instruments [3,4], is well suited to turning such signals into sound; CsoundUnity [5] runs it alongside real-time Unity visuals. Cabbage [6] provides Csound instruments with graphical controls whose header declarations can also describe parameters to an external system.

Building an EEG-to-Csound system raises two related but distinct problems, one at design time and one at run time. At design time, creators must decide which EEG features should drive which synthesis parameters, a search that is normally done by hand, through repeated inspection and testing. At run time, the system must track continuous EEG changes through immediate, moment-to-moment adjustments while also making occasional, coordinated changes that take the current state and the audiovisual scene as a whole into account. Our design assigns each problem to a different component.

We address the first problem with a mapping assistant that turns natural-language intent and automatically discovered parameter metadata into editable binding suggestions. A scene agent addresses the second by returning sound and visual actions for configured runtime events. These roles separate mapping design from runtime audiovisual behavior. The two contributions are an LLM-assisted,

human-in-the-loop mapping workflow and a two-path runtime architecture combining continuous EEG modulation with discrete scene-agent actions.

2 System Architecture

The system forms a real-time audiovisual biofeedback loop (Fig. 1). The listener’s EEG drives sound and a synchronized visual scene while acquisition continues as the listener experiences the output.

NeurOSC currently acquires data from a Muse S Athena headset through the Muse software development kit (SDK; v8.0.5) [7]. A second backend for OpenBCI Ganglion and related boards [8], built on BrainFlow [9], is under development and is not part of the currently tested pipeline. NeurOSC provides device-adapted digital signal processing (DSP) presets whose individual steps remain configurable. For the Muse pipeline, these steps include reference handling, detrending, 1–45 Hz band-pass filtering, and 50/60 Hz notch filtering. Signal-quality and electrode-contact diagnostics support operator monitoring but do not automatically gate feature extraction.

From four-second windows with 50% overlap, Welch power spectral density analysis estimates delta, theta, alpha, beta, and gamma power. NeurOSC computes their cross-channel mean relative composition and broadcasts the resulting 0–1 features over OSC [10] at approximately 10 Hz.

NeurOSC makes these features available in several representations for different mapping needs. The demonstration uses the cross-channel mean relative band composition. NeurOSC can also send an optional baseline-change representation, calculated in decibels relative to a fixed personal session baseline, allowing a downstream mapping to respond to increases or decreases relative to the listener’s reference; this representation is not currently consumed by Unity. Unity instead performs its own 90-second calibration at the start of each session, converting subsequent values into Z-scores and classifying them with a heuristic classifier into one of five system-defined labels: *Focused*, *Relaxed*, *Aroused*, *Drowsy*, or *Neutral*.

The system uses LLMs in two stages. Before a session, the mapping assistant generates and auditions EEG-to-Csound bindings. During a session, the scene agent turns confirmed state changes and other configured events into sound and visual actions using scene context and the creator’s creative direction. A configurable cloud application programming interface (API) layer supports OpenAI [11], Anthropic [12], Gemini [13], and OpenAI-compatible services such as DeepSeek [14]; the configuration reported here uses OpenAI. Both assistants draw on the same parameter catalog, although each receives the subset required for its task.

At run time, the continuous and discrete paths operate together. CsoundUnity passes their parameter updates to the running Csound instrument, while Unity renders and controls the visual scene. Section 3.2 explains how the paths combine when they address the same parameter.

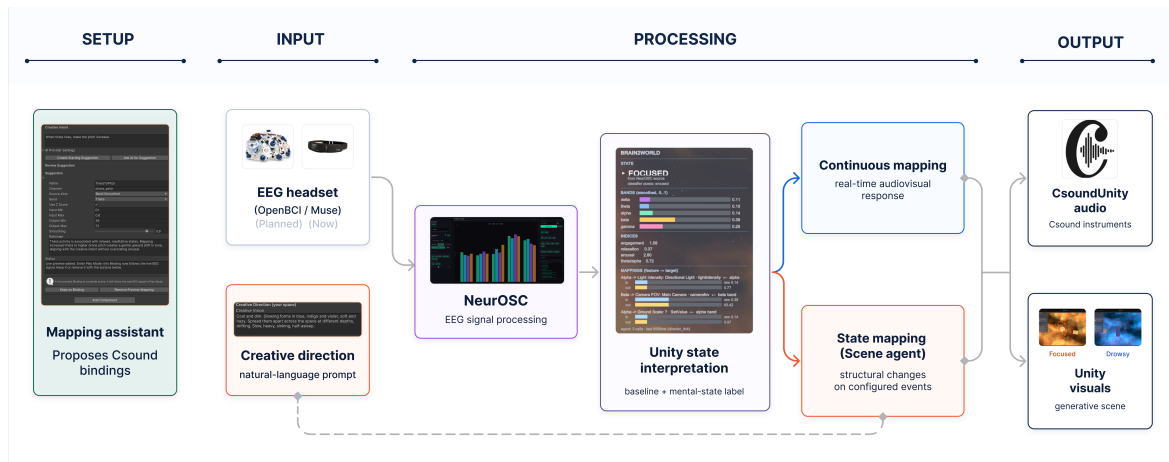


Fig. 1. Brain2World signal and control pipeline.

3 EEG-to-Csound Mapping

3.1 Parameter Auto-Discovery

Each instrument declares its controls in its Cabbage header. For example, “brain_drone” declares a control channel named “filter_cutoff”, with a range from 200 to 8000, a default value of 1200, and the label “cutoff Hz” (Listing 1). At load time, the system parses these declarations into a shared parameter catalog. Agent-generated targets for recognized channels are clamped to their catalogued ranges before execution; an unrecognized channel name produces a warning rather than being strictly rejected.

The mapping assistant uses each channel’s name, range, and description to propose bindings. The runtime scene agent prompt also includes default values and the instrument name. Because it does not include current parameter values, defaults provide reference points for selecting new targets, while the instrument name identifies the control set being addressed.

In our demonstration, “brain_drone” exposes six controls through this mechanism. A second instrument, “brain_fm”, uses a different Cabbage header and is discovered in the same way. Assigning a different instrument and rebuilding the catalog supplies its new control set automatically, without registering each parameter individually in Unity.

```
<Cabbage>
rslider channel("master_level"),    range(0,1,0.7)    text("overall gain")
rslider channel("drone_pitch"),      range(36,72,48)   text("root MIDI note")
rslider channel("filter_cutoff"),    range(200,8000,1200) text("cutoff Hz")
rslider channel("brightness"),       range(0,1,0.3)    text("overtones")
rslider channel("texture_density"),  range(0,1,0.2)    text("noise shimmer")
rslider channel("reverb_mix"),       range(0,1,0.4)    text("reverb")
</Cabbage>
```

Listing 1. A Cabbage header declares an instrument’s parameter set.

3.2 Two-Path Mapping

The system drives Csound through two runtime paths that differ in when they activate and what information they use. The continuous path runs every frame. Each configured binding connects an EEG feature to a Csound channel. It reads the selected feature, optionally converts it to a Z-score using Unity’s session calibration reference, maps the result through a shaping curve to the channel’s output range, and smooths the value before writing it. Each binding has its own smoothing setting. Table 1 lists the six continuous bindings active in our demonstration instrument, “brain_drone”.

Table 1. Continuous EEG-to-Csound bindings for “brain_drone”.

Csound parameter	Driven by (EEG feature)
“reverb_mix”	alpha (relaxation)
“brightness”	gamma and arousal
“filter_cutoff”	engagement (beta)
“texture_density”	beta
“drone_pitch”	theta (inverse)
“master_level”	arousal

The discrete path activates on events rather than every frame. The scene agent may be triggered by a confirmed state transition, an EEG feature event such as a band surge or change in dominant band, or a periodic director tick that requests an incremental scene update. A state transition is confirmed after a candidate label remains unchanged for a configurable dwell period, set to two seconds here. Each request includes the trigger type, current and previous states, the current Z-scored feature bundle, recent event history, scene context, creative direction, and parameter catalog; the agent returns sound and visual actions.

The two paths combine at the parameter level. The scene agent sets larger-scale base targets for selected Csound parameters, and each target glides toward its new value. Continuous bindings

provide smaller-scale modulation around these targets. For every shared parameter, the final value is the current base target plus continuous modulation, clamped to the parameter range. Discrete events shape the broader audiovisual structure, while ongoing EEG activity adds variation without overriding the agent’s targets.

To produce these state transitions, Unity derives three composite features, labelled engagement, relaxation, and arousal, from the smoothed relative band values. The session calibration converts these features, the theta/alpha ratio, and individual band values into Z-scores. The heuristic classifier then applies the rules in Table 2. When several conditions hold, it evaluates them in the order *Aroused*, *Focused*, *Relaxed*, and *Drowsy*; otherwise it assigns *Neutral*. The table also lists the audiovisual guidance included in the scene agent prompt. These choices draw on prior work relating structural musical features to emotional expression [15].

Table 2. System-defined state labels, classifier conditions, and audiovisual guidance.

State	Classifier condition	Audiovisual guidance
Focused	Engagement Z-score > 1 and theta/alpha Z-score < 0	Structured and clear; brighter sound with lower reverb
Relaxed	Relaxation Z-score > 1 and arousal Z-score < 0	Flowing and spacious; gentle filtering and increased reverb
Aroused	Arousal Z-score > 1 or gamma Z-score > 1.5	Bright, active, and dense
Drowsy	Theta Z-score > 1 and beta Z-score < 0	Slow, low, and dark
Neutral	No other condition is met	Minimal visual movement and a steady sound

4 Discussion and Future Work

The prototype establishes a real-time path from Muse EEG acquisition to coordinated sound and visual output, with a simulator for development without hardware. The next device-level step is to extend the Muse pipeline to OpenBCI through BrainFlow. Signal handling will be improved by using the existing quality and contact diagnostics to identify and exclude segments affected by motion, poor contact, or environmental noise.

User studies will examine how creators generate, audition, revise, and retain mapping suggestions across instruments and sessions. A separate listening study will test whether state transitions remain perceptually clear as the scene agent varies its sound and visual actions. The resulting observations can then guide revisions to the agent’s prompts, state descriptions, and action constraints.

Longer-term work will focus on creative use. Pairing the system with Dr.C [16], a human-in-the-loop agent for Csound sound design, would allow creators to design instruments from musical descriptions and pass their controls directly into the mapping workflow. Further directions include longer scene memory, virtual reality, and multi-speaker presentation. The current prototype provides a basis for composition, performance, and immersive installations in which listeners’ brain activity becomes part of the creative material.

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